Deformed characteristic classes

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1 The starting point

Given a vector bundle E over a smooth manifold M, taking characteristic classes of E, we get cohomology classes of the base manifold M that give us a way of classifying vector bundles that can be defined on M. We want to generalize the Chern Weil homomorphism in such a way that taking characteristic classes of the bundle yields us cohomology classes in the deformed or twisted cohomology group of the base manifold M. Recall that for a closed 1-form $\psi \in \Omega^1(M)$ on M, if we *deform* or *twist* the exterior derivative $d : \Omega^k(M) \to \Omega^{k+1}(M)$, via imposing that

$$d_\psi := d + \psi\wedge$$

where the closedness of ψ implies that $d_\psi^2 = 0$ because

$$\begin{aligned} d_\psi d_\psi(f) &= d_\psi(df + \psi f) \\ &= (d + \psi\wedge)(df + \psi f) \\ &= d^2 f + d(\psi f) + \psi \wedge df + \psi \wedge \psi f \\ &= 0 + d\psi \wedge f - \psi \wedge df + \psi \wedge df + 0 \\ &= 0 \end{aligned}$$

that let's us define the cohomology

$$H^k_\psi(M) := \frac{Ker(d^k_\psi : \Omega^k(M) \to \Omega^{k+1}(M))}{Im(d^{k-1}_\psi : \Omega^{k-1}(M) \to \Omega^k(M))}$$

Take for example the Chern classes, for which the twisting parameter $\psi \in \Omega(M, \mathbf{C})$ must be complex-valued and the vector bundle should be complexified, then our twisted Chern classes should be something of the sort

$$c_{i,\psi}(E) \in H^{2i}_\psi(M)$$

such that if we annihilate ψ i.e. set $\psi = 0$, then everything becomes as it were in the classical theory i.e. $c_{i,\psi=0}(E) = c_i(E) \in H^{2i}(M) = H^{2i}_{\psi=0}(M)$.

Recall that a connection on the vector bundle E is an endomorphism on the space of sections of E

$$\nabla_X : \Gamma(M, E) \to \Gamma(M, E)$$

where X is vector field on M, that satisfies

$$\nabla_X(fs) = X(f)s + f\nabla_X s \tag{1.1}$$

where $s \in \Gamma(M, E)$ and $f \in C^\infty(M)$.
For a local frame e_i over $U \subset M$, the sections $s \in \Gamma(U, E)$ are expressed as $s = \sum_i h^i e_i$, for which

$$\nabla_X e_j = \omega_j^i e_i$$

(where we have used Einstein's summation convention).

Definition 1.1. We define the connection

$$\nabla_{X,\psi}(fs) = X_\psi(f)s + f\nabla_{X,\psi}(s) \tag{1.2}$$

where the vector field

$$X_\psi(f) = X(f) + \psi(X).f$$

And for the frames e_j

$$\nabla_{X,\psi}(e_j) = \omega_j^i(X)e_i + \delta_j^i\psi(X)e_i \tag{1.3}$$

which is essentially a connection on the tensor product bundle $E \otimes \underline{\mathbf{R}}$ where $\underline{\mathbf{R}}$ is the trivial line bundle with the connection d_ψ. Let check does it follow $\nabla_{fX,\psi} = f\nabla_{X,\psi}$?

$$\begin{aligned}
\nabla_{fX,\psi}e_j &= \omega_j^i(fX)e_i + \delta_j^i\psi(fX)e_i \\
&= f\omega_j^i(X)e_i + f\delta_j^i\psi(X)e_i \\
&= f(\omega_j^i(X)e_i + \delta_j^i\psi(X)e_i) \\
&= f\nabla_{X,\psi}
\end{aligned}$$

Okay, let's check the bi-linearity of our connection,

$$\begin{aligned}
\nabla_{X+Y,\psi}e_j &= \nabla_{X+Y}e_j + \delta_j^i\psi(X+Y)e_i \\
&= \nabla_{X,\psi}e_j + \nabla_{Y,\psi}e_i + \delta_j^i\psi(X)e_i + \delta_j^i\psi(Y)e_i \\
&= \nabla_{X,\psi}e_j + \delta_j^i\psi(X)e_i + \nabla_{Y,\psi}e_j + \delta_j^i\psi(Y)e_i \\
&= (\nabla_{X,\psi} + \nabla_{Y,\psi})e_j
\end{aligned}$$

Given a field of covectors (or one-form) α defined in a neighborhood of p, it's covariant derivative $(\nabla_{\mathbf{X},\psi}\alpha)_p$ is defined in a way to make the resulting operation compatible with tensor contraction and the product rule. That is, $(\nabla_{\mathbf{X},\psi}\alpha)_p$ is

defined as the unique one-form at p such that the following identity is satisfied for all vector fields u in a neighborhood of p,

$$\left(\nabla_{\mathbf{X},\psi}\alpha\right)_p\left(\mathbf{u}_p\right) \stackrel{?}{=} \nabla_{\mathbf{X},\psi}\left[\alpha\left(\mathbf{u}\right)\right]_p - \alpha_p\left[\left(\nabla_{\mathbf{X},\psi}\mathbf{u}\right)_p\right]$$

The covariant derivative of a covector field along a vector field X should be again a covector field. We need this because we want similar gadgetry for form also [**Need to check the definition**].

Q: Can we extend it for any tensor field also?

Let's derive the twisted derivative of the wedge product,

$$\begin{aligned}
d_\psi(\alpha\wedge\beta) &= d(\alpha\wedge\beta) + \psi\wedge\alpha\wedge\beta \\
&= d\alpha\wedge\beta + (-1)^{|\alpha|}\alpha\wedge d\beta + \psi\wedge\alpha\wedge\beta \\
&= (d\alpha + \psi\wedge\alpha\wedge\beta) + (-1)^{|\alpha|}\alpha\wedge d\beta \\
&= (d\alpha + \psi\wedge\alpha)\wedge\beta + (-1)^{|\alpha|}\alpha\wedge d\beta \\
&= d_\psi\alpha\wedge\beta + (-1)^{|\alpha|}\alpha\wedge d\beta \\
&= d_\psi\alpha\wedge\beta + (-1)^{|\alpha|}\alpha\wedge(d_\psi\beta - \psi\wedge\beta) \\
&= d_\psi\alpha\wedge\beta + (-1)^{|\alpha|}\alpha\wedge d_\psi\beta + (-1)^{|\alpha|+1}\alpha\wedge\psi\wedge\beta
\end{aligned}$$

Example 1.1. Let's compute the twisted cohomology group $H^\bullet_\omega(M)$. First, observe that, since $\omega\in\Omega^1(M), \omega = h(x)dx$ for $h(x)\in C^\infty(M)$. Lift $h(x)$ to a periodic function on $\mathbb{R}$.

Q: Then if $[\omega]=0$, what can we say about $\int_0^1 h(x)dx = ?$.
Now,

$$\ker\left(d_\omega : \Omega^0(M)\to\Omega^1(M)\right) = \{f\in C^\infty(M) \mid d_\omega f = 0\}$$

The condition $d_\omega f = 0$ says that $f'(x) = -h(x)f(x)$

$$\begin{aligned}
f'(x) &= -h(x)f(x) \\
\frac{f'}{f} &= -h \\
\ln f &= \int -h(x)dx + C \\
f(x) &= e^C e^{\int -h(x)dx} \\
f(x) &= Ae^{\int -h(x)dx}
\end{aligned}$$

But if $f\in C^\infty(M)$ then $f(1)=f(0)$ so

$$e^{-\int_0^1 h(x)dx} = 1 \Longrightarrow \int_0^1 h(x)dx = 0$$

Therefore, if ω is exact,

$$\ker d_\omega = \left\{ Ae^{(-\int h(x)dx)} \middle| A \in \mathbb{R} \right\}$$

and $H^0_\omega(M) \cong \mathbb{R}$;

Q: and if ω is not exact, $\ker d_\omega \equiv ?$ and $H^0_\omega(M) \cong ?$.

If $\alpha \in \operatorname{im} d_\omega : \Omega^0(M) \to \Omega^1(M)$, then for some $f \in C^\infty(M), \alpha = d_\omega f$. Additionally, since $M = \mathbb{R}/\mathbb{Z}$, we can consider behavior on the universal cover $\widetilde{M} = \mathbb{R}$. Since $\alpha \in \Omega^1(M)$, we can denote $\alpha = g(x)dx$ for some $g \in C^\infty(\mathbb{R})$. So

$$\begin{aligned} d_\omega f = gdx &\iff \\ (d+\omega)f = gdx &\iff \\ f' + hf = g & \end{aligned}$$

Solving the first order linear differential equation, we see that

$$f(x) = e^{-\int h dx} \int g \cdot e^{\int h dx} dx + Ce^{-\int h dx}$$

Letting $H(x) = e^{\int h(x)dx}$, we see that

$$H(0) = e^{\int_0^0 h(x)dx} = 1$$

and

$$f(x) = \frac{1}{H(x)} \left[\int_0^x g(t) \cdot H(t)dt + C \right]$$

Note that

$$f(0) = \frac{1}{H(0)} \left[\int_0^0 g(t) \cdot H(t)dt + C \right] = C$$

So

$$f(x) = \frac{1}{H(x)} \left[\int_0^x g(t) \cdot H(t)dt + f(0) \right] \tag{1.4}$$

So α is exact precisely when a solution to 1.4 descends to M; in particular, if 1.4 has a periodic solution. It is then necessary to consider two cases:
Case I: $[\omega] = 0$ (ω is exact) : Again, $\int_0^1 h(x)dx = 0$ so $H(1) = H(0) = 1$ so, from 1.4, we have

$$\begin{aligned} f(x) &= \frac{1}{H(x)} \left[\int_0^x g(t) \cdot H(t)dt + f(0) \right] \quad \text{and} \\ f(1) &= \int_0^1 g(t) \cdot H(t)dt + f(0) \end{aligned}$$

Since $f(x) \in C^\infty(M), f(1) = f(0)$, so α is exact with respect to the twisted differential on M if and only if $\int_0^1 g(x) \cdot H(x)dx = 0$. So, $H^1_\omega \cong \mathbb{R}$. - Case II: $[\omega] \neq 0$ (ω is not exact) :

If ω is not exact, then $H(1) = A \neq 1 = H(0)$.

$$f(x) = \frac{1}{H(x)} \left[\int_0^x g(t) \cdot H(t)dt + f(0) \right] \quad \text{and}$$
$$f(1) = \frac{1}{A} \left[\int_0^1 g(t) \cdot H(t)dt + f(0) \right]$$

Again, since $f(0) - f(1) = 0$ and $f \in C^\infty(M)$,

$$f(0) = \frac{1}{A} \left[\int_0^1 g(x) \cdot H(x) + f(0) \right] dx \iff$$
$$f(0) = \frac{1}{A-1} \int_0^1 g(x) \cdot H(x)dx$$

So there is exactly one periodic solution and α is exact with respect to the twisted differential.

Hence, if ω is exact,

$$H^\bullet_\omega = \begin{cases} \mathbb{R} & \bullet = 0, 1 \\ 0 & \text{otherwise} \end{cases}$$

If ω is not exact, then

$$H^\bullet_\omega = 0$$

which agrees with the usual cohomology with local coefficients of S^1.

Example 1.2. Let's compute the twisted cohomology of the 2-dimensional torus T^2. The torus T^2 is the product $S^1 \times S^1$, with coordinates (x_1, x_2), where each $x_i \in \mathbb{R}/\mathbb{Z}$ (so x_1, x_2 are periodic coordinates). We twist the exterior differential operator d by a 1-form $\omega = h_1(x_1, x_2)dx_1 + h_2(x_1, x_2)dx_2$, where $h_1(x_1, x_2)$ and $h_2(x_1, x_2)$ are smooth functions on T^2. The twisted exterior differential is given by:

$$d_\omega = d + \omega\wedge$$

Our goal is to compute the twisted cohomology groups $H^k_\omega(T^2)$, for $k = 0, 1, 2$, i.e., we want to compute the cohomology of the twisted complex:

$$\Omega^0(T^2) \xrightarrow{d_\omega} \Omega^1(T^2) \xrightarrow{d_\omega} \Omega^2(T^2)$$

where $\Omega^k(T^2)$ denotes the space of smooth differential k-forms on T^2.

Let $f(x_1, x_2) \in \Omega^0(T^2)$ be a smooth function. The twisted differential acts as:

$$d_\omega f = df + \omega \wedge f = \frac{\partial f}{\partial x_1} dx_1 + \frac{\partial f}{\partial x_2} dx_2 + h_1 f dx_1 + h_2 f dx_2$$

So the equation $d_\omega f = 0$ (for $f \in \ker d_\omega$) becomes:

$$\frac{\partial f}{\partial x_1} + h_1 f = 0, \quad \frac{\partial f}{\partial x_2} + h_2 f = 0$$

These are two decoupled first-order partial differential equations for $f(x_1, x_2)$. The general solution is:

$$f(x_1, x_2) \stackrel{?}{=} C \cdot \exp(?)$$

where $C \in \mathbb{R}$ is a constant, and the integrals are understood to be indefinite with respect to x_1 and x_2. If h_1 and h_2 are periodic, then $f(x_1, x_2)$ is a periodic function on the torus.

Thus, the space of functions satisfying $d_\omega f = 0$ is one-dimensional (up to a constant multiple of the solution), so:

$$H^0_\omega(T^2) \cong \mathbb{R}$$

Next, we consider the space of 1-forms $\alpha \in \Omega^1(T^2)$. A general 1-form is:

$$\alpha = a(x_1, x_2)dx_1 + b(x_1, x_2)dx_2$$

The twisted differential d_ω acts on 1-forms as:

$$d_\omega \alpha = d\alpha + \omega \wedge \alpha$$

First, compute the exterior derivative of α:

$$d\alpha = \left(\frac{\partial b}{\partial x_1} - \frac{\partial a}{\partial x_2} \right) dx_1 \wedge dx_2$$

and the wedge product $\omega \wedge \alpha$ gives:

$$\omega \wedge \alpha = (h_1 a + h_2 b) dx_1 \wedge dx_2$$

Thus, the equation $d_\omega \alpha = 0$ becomes:

$$\frac{\partial b}{\partial x_1} - \frac{\partial a}{\partial x_2} + h_1 a + h_2 b = 0$$

This is a single first-order partial differential equation in two variables. The space of solutions to this equation is more complicated to describe explicitly, but it's known that on T^2, the twisted cohomology in degree 1 is two-dimensional. This corresponds to the fact that the torus has two nontrivial cycles (in the x_1- and x_2-directions), and twisting by ω introduces a deformation that affects both cycles.

Thus, we conclude:

$$H^1_\omega(T^2) \cong \mathbb{R}^2 \quad \text{Is it okay?}$$

Finally, consider a general 2-form $\beta \in \Omega^2(T^2)$:

$$\beta = c(x_1, x_2)dx_1 \wedge dx_2$$

The twisted differential acts as:

$$d_\omega \beta = d\beta + \omega \wedge \beta$$

Since $d\beta = 0$ (there are no 3-forms on T^2), we only have:

$$d_\omega \beta = \omega \wedge \beta = 0$$

because $\omega \wedge \beta = 0$ automatically for any 2-form β on a 2-dimensional manifold. Thus, all 2-forms are in the kernel of d_ω, so:

$$H^2_\omega(T^2) \cong \mathbb{R}$$

The twisted cohomology groups of the 2-torus T^2 with the twist given by $\omega = h_1(x_1, x_2)dx_1 + h_2(x_1, x_2)dx_2$ are:

$$H^0_\omega(T^2) \cong \mathbb{R}, \quad H^1_\omega(T^2) \cong \mathbb{R}^2, \quad H^2_\omega(T^2) \cong \mathbb{R}$$

This is the same as the untwisted cohomology for T^2.

1.1 Twisted Curvature

So, far we have:

$$\nabla_{Y,\psi}(e_j) = \omega_j^k(Y)e_k + \delta_j^k \psi(Y)e_k$$

Now, let's try to find the notion of curvature:

$$\begin{aligned}
&\nabla_{X,\psi}\nabla_{Y,\psi}e_j \\
&= \nabla_{X,\psi}(\omega_j^k(Y)e_k + \delta_j^k\psi(Y)e_k) \\
&= X_\psi(\omega_j^k(Y))e_k + \omega_j^k(Y)\nabla_{X,\psi}e_k + X_\psi(\delta_j^k\psi(Y))e_k \\
&\qquad\qquad + \delta_j^k\psi(Y)\nabla_{X,\psi}e_k \\
&= (X(\omega_j^k(Y)) + \psi(X)\omega_j^k(Y))e_k \\
&\qquad + \omega_j^k(Y)(\omega_k^i(X)e_i + \delta_k^i\psi(X)e_i) \\
&\qquad + (X(\delta_j^k\psi(Y)) + \psi(X)\delta_j^k\psi(Y))e_k \\
&\qquad + \delta_j^k\psi(Y)(\omega_k^i(X)e_i + \delta_k^i\psi(X)e_i)
\end{aligned} \tag{1.5}$$

And

$$\begin{aligned}
\nabla_{Y,\psi}\nabla_{X,\psi}e_j = &(Y(\omega_j^k(X)) + \psi(Y)\omega_j^k(X))e_k \\
&+ \omega_j^k(X)(\omega_k^i(Y)e_i + \delta_k^i\psi(Y)e_i) \\
&+ (Y(\delta_j^k\psi(X)) + \psi(Y)\delta_j^k\psi(X))e_k \\
&+ \delta_j^k\psi(X)(\omega_k^i(Y)e_i + \delta_k^i\psi(Y)e_i)
\end{aligned} \tag{1.6}$$

and we have that

$$\nabla_{[X,Y],\psi}e_j = \omega_j^i([X,Y])e_i + \delta_j^i\psi([X,Y])e_i$$

Thus,

$$\begin{aligned} R_\psi(X,Y)e_j &= \nabla_{X,\psi}\nabla_{Y,\psi}e_j - \nabla_{Y,\psi}\nabla_{X,\psi}e_j - \nabla_{[X,Y],\psi}e_j \\ &= [X(\omega(Y)) - Y(\omega(X)) - \omega([X,Y]) + X(\psi(Y)) - Y(\psi(X)) - \psi([X,Y]) \\ &\quad \omega\wedge\omega(X,Y) + \psi\wedge\omega(X,Y)]e_j \\ &= d_\psi\omega + \omega\wedge\omega + X(\psi(Y)) - Y(\psi(X)) - \psi([X,Y]) \end{aligned} \tag{1.7}$$

If we choose vector fields X, Y that satisfy the condition

$$\psi([X,Y]) = X(\psi(Y)) - Y(\psi(X)) \tag{1.8}$$

then in the last equation, only the first two terms remain. So we have defined the twisted curvature 2-form

$$\Omega_\psi = d_\psi\omega + \omega\wedge\omega \tag{1.9}$$

that acts on vector fields satisfying condition (4)[fix labelling?]. Vector fields satisfying this condition will be called ψ-variant vector fields.
The twisted curvature satisfies the second Bianchi identity. Indeed,

$$\begin{aligned} d_\psi\Omega_\psi &= d_\psi(d_\psi\omega) + (d_\psi\omega)\wedge\omega - \omega\wedge(d_\psi\omega) \\ &= (\Omega_\psi - \omega\wedge\omega)\wedge\omega - \omega\wedge(\Omega_\psi - \omega\wedge\omega) \\ &= \Omega_\psi\wedge\omega - \omega\wedge\Omega_\psi \end{aligned} \tag{1.10}$$

And a generalized form of that:

$$d_\psi\Omega_\psi^k = \Omega_\psi^k\wedge\omega - \omega\wedge\Omega_\psi^k \tag{1.11}$$

Proof. We can prove it using induction technique. We already proved the base case $k = 1$. Let's assume it's true for $k-1$,

$$d_\psi\Omega_\psi^{k-1} = \Omega_\psi^{k-1}\wedge\omega - \omega\wedge\Omega_\psi^{k-1}.$$

Then for k we have,

$$\begin{aligned} d_\psi\Omega_\psi^k &= d_\psi(\Omega_\psi^{k-1}\wedge\Omega_\psi) \\ &= d_\psi\Omega_\psi^{k-1}\wedge\Omega_\psi + \Omega_\psi^{k-1}\wedge d_\psi\Omega_\psi \qquad (\deg\Omega_\psi^{k-1} = 2k-2) \\ &= (\Omega_\psi^{k-1}\wedge\omega - \omega\wedge\Omega_\psi^{k-1})\wedge\Omega_\psi + \Omega_\psi^{k-1}\wedge(\Omega_\psi\wedge\omega - \omega\wedge\Omega_\psi) \\ &= \Omega_\psi^{k-1}\wedge\omega\wedge\Omega_\psi - \omega\wedge\Omega_\psi^k + \Omega_\psi^k\wedge\omega - \Omega_\psi^{k-1}\wedge\omega\wedge\Omega_\psi \\ &= \Omega_\psi^k\wedge\omega - \omega\wedge\Omega_\psi^k \end{aligned}$$

□

2 Chern Weil homomorphism

Consider a smooth family of twisted connections (i.e. connections given by (2))

$$\nabla^t_{X,\psi} = (1-t)\nabla^0_{X,\psi} + t\nabla^1_{X,\psi} \tag{2.1}$$

where we are assuming that all the connections in this t-smooth family are twisted by the same twisting parameter ψ; the interesting and difficult case where there is a t-smooth family of twisting parameters $\{\psi_t\}$ is left for future work.
For such a family, there is a family of twisted curvature $\Omega_{\psi,t}$.
We will use the dotted notation to mean derivative with respect to the parameter t:

$$\begin{aligned}\dot{\Omega}_{\psi,t} &= \frac{d}{dt}\Omega_{\psi,t} \\ &= \frac{d}{dt}d_\psi\omega_t + \frac{d}{dt}\omega_t\wedge\omega_t + \omega_t\wedge\frac{d}{dt}\omega_t \\ &= d_\psi\dot{\omega}_t + \dot{\omega}_t\wedge\omega_t + \omega_t\wedge\dot{\omega}_t\end{aligned}$$

Now,

$$\begin{aligned}\frac{d}{dt}(\operatorname{Tr}\Omega^k_{\psi,t}) &= \operatorname{Tr}(\frac{d}{dt}\Omega^k_{\psi,t}) \\ &= \operatorname{Tr}(\dot{\Omega}_{\psi,t}\wedge\Omega^{k-1}_{\psi,t} + \Omega_{\psi,t}\wedge\dot{\Omega}_{\psi,t}\wedge\Omega^{k-2}_{\psi,t} + ... + \Omega^{k-1}_{\psi,t}\wedge\dot{\Omega}_{\psi,t}) \\ &= k\operatorname{Tr}(\Omega^{k-1}_{\psi,t}\wedge\dot{\Omega}_{\psi,t}) \\ &= k\operatorname{Tr}(\Omega^{k-1}_{\psi,t}\wedge[d_\psi\dot{\omega} + \dot{\omega}\wedge\omega + \omega\wedge\dot{\omega}]) \\ &= k\operatorname{Tr}(\Omega^{k-1}_{\psi,t}\wedge d_\psi\dot{\omega} - \omega\wedge\Omega^{k-1}_{\psi,t}\wedge\dot{\omega} + \Omega^{k-1}_{\psi,t}\wedge\omega\dot{\omega}) \\ &= k\operatorname{Tr}(\Omega^{k-1}_{\psi,t}\wedge d_\psi\dot{\omega} + [\Omega^{k-1}_{\psi,t}\wedge\omega - \omega\wedge\Omega^{k-1}_{\psi,t}]\wedge\dot{\omega}) \\ &= k\operatorname{Tr}(\Omega^{k-1}_{\psi,t}\wedge d_\psi\dot{\omega} + d_\psi\Omega^{k-1}_\psi\wedge\dot{\omega})\end{aligned}$$

Now, we know that,

$$d_\psi(\Omega^{k-1}_\psi\wedge\dot{\omega}) = d_\psi\Omega^{k-1}_\psi\wedge\dot{\omega} + \Omega^{k-1}_\psi\wedge d_\psi\dot{\omega} - \Omega^{k-1}_\psi\wedge\psi\wedge\dot{\omega}$$
$$d_\psi(\Omega^{k-1}_\psi\wedge\dot{\omega}) + \Omega^{k-1}_\psi\wedge\psi\wedge\dot{\omega} = \Omega^{k-1}_\psi\wedge d_\psi\dot{\omega} + d_\psi\Omega^{k-1}_\psi\wedge\dot{\omega}$$

Then,

$$\begin{aligned}\operatorname{Tr}(\Omega^{k-1}_{\psi,t}\wedge d_\psi\wedge\dot{\omega} + d_\psi\Omega^{k-1}_\psi\wedge\dot{\omega}) &= \operatorname{Tr}(d_\psi(\Omega^{k-1}_\psi\wedge\dot{\omega} + \Omega^{k-1}_\psi\wedge\psi\wedge\dot{\omega}) \\ &= \operatorname{Tr}(d_\psi(\Omega^{k-1}_\psi\wedge\dot{\omega}) + \operatorname{Tr}(\Omega^{k-1}_\psi\wedge\psi\wedge\dot{\omega})\end{aligned}$$

Now, if we define our curvature traceless $2k$ form then,

$$\operatorname{Tr}(\Omega^{k-1}_{\psi,t}\wedge d_\psi\wedge\dot{\omega} + d_\psi\Omega^{k-1}_\psi\wedge\dot{\omega}) = d_\psi(\operatorname{Tr}(\Omega^{k-1}_\psi\wedge\dot{\omega}))$$

$$\frac{d}{dt}(\operatorname{Tr}\Omega_{\psi,t}^k) = d_\psi(k\operatorname{Tr}(\Omega_\psi^{k-1}\wedge\dot{\omega}))$$
$$\int_{t=0}^1 \frac{d}{dt}(\operatorname{Tr}\Omega_{\psi,t}^k)dt = \int_0^1 d_\psi(k\operatorname{Tr}(\Omega_\psi^{k-1}\wedge\dot{\omega}))dt$$
$$\operatorname{Tr}(\Omega_{\psi,1}^k) - \operatorname{Tr}(\Omega_{\psi,0}^k) = d_\psi\int_0^1 (k\operatorname{Tr}(\Omega_\psi^{k-1}\wedge\dot{\omega}))dt$$
$$= d_\psi\Phi^{k-1}$$

We need to show that,

$$d_\psi(\Omega_\psi^{k-1}\wedge\omega) = d_\psi\Omega_\psi^{k-1} + \Omega_\psi^{k-1}\wedge d_\psi\omega$$

Let's back to our discussion,

$$d_\psi(\Omega^{k-1}\wedge\omega) = d_\psi\Omega_\psi^{k-1}\wedge\omega + \Omega_\psi^{k-1}\wedge d_\psi\omega + (-1)^{2k-1}\Omega_\psi^{k-1}\wedge\psi\wedge\omega$$
$$= d_\psi\Omega_\psi^{k-1}\wedge\omega + +\Omega_\psi^{k-1}\wedge d_\psi\omega - \Omega_\psi^{k-1}\wedge\psi\wedge\omega$$

Integrating over the parameter t yields,

$$\int_0^1 \frac{\mathrm{d}}{\mathrm{d}t}(\operatorname{Tr}(\Omega_{\psi,\mathrm{t}}^{\mathrm{k}}))dt = \int_0^1 kd_\psi(\operatorname{Tr}(\Omega_\psi^{k-1}\wedge\dot{\omega}_t))$$
$$\operatorname{Tr}(\Omega_{\psi,1}^k) - \operatorname{Tr}(\Omega_{\psi,0}^k) = d_\psi\int_0^1 k(\operatorname{Tr}(\Omega_\psi^{k-1}\wedge\dot{\omega}_t))$$

So the cohomology class (which is an element of $H_\psi^*(M)$) that is represented by the form $\operatorname{Tr}(\Omega_\psi^k)$ is independent of the twisted connection.
The generalized Bianchi second identity (10) also implies that the form $\operatorname{Tr}(\Omega_\psi^k)$ is closed:

$$d_\psi(\operatorname{Tr}(\Omega_\psi^k)) = \operatorname{Tr}(d_\psi\Omega_\psi^k)$$
$$\implies \operatorname{Tr}(\Omega_\psi^k\wedge\omega - \omega\wedge\Omega_\psi^k)$$
$$= \quad 0$$

So the form $\operatorname{Tr}(\Omega_\psi^k)$ defines a cohomology class in in the twisted cohomology $H_\psi^*(M)$.

Let's define Chern class for our Ω_ψ by,

$$\det\left(I + \frac{i}{2\pi}\Omega_\psi\right) = 1 + c_{1,\psi}(E) + c_{2,\psi}(E) + \cdots$$

these Chern classes will be called *twisted* Chern classes. Mimicking the classical Chern class theory, we defined the *Deformed Calabi Yau* manifold as a compact Kähler manifold, whose first twisted Chern class vanishes. Cohomologically deforming it, we speculate that converging the complex valued smooth function

$\psi(x)$ to zero, the twisted Chern classes reduces to ordinary Chern classes, hence the *deformed* Calabi Yau manifold reduces to Calabi Yau manifolds. So we state the following:
Conjecture: For every $[\psi] \in H^1_{dR}(M, \mathbb{C})$, there exists a compact Kähler manifold $\tilde{M}$ such that $c_{1,\psi}(\tilde{M}) = 0$ then we can take the limit $\psi \to 0$ if and only if $\tilde{M}$ is a Calabi-Yau manifold.
Here ψ is a complex-valued differential form.

3 Complex setting

Let ψ be a closed complex valued 1-form on a complex manifold M (in this section M will be a compact complex manifold of complex dimension n). From the (fixed) complex structure of M we have the splitting

$$\psi = \Psi^{1,0}(x^i)dx^i + \Psi^{0,1}(\overline{x}^i)d\overline{x}^i$$

but we will briefly write that as

$$\psi = \psi^{1,0} + \psi^{0,1}$$

where $\psi^{1,0} \in \Omega^{1,0}(M)$ and $\psi^{0,1} \in \Omega^{0,1}(M)$
So the *twisted* derivative now reads

$$\begin{aligned} d_\psi &= d + \psi\wedge \\ &= (\partial + \overline{\partial}) + (\psi^{1,0} + \psi^{0,1})\wedge \\ &= (\partial + \psi^{1,0}\wedge) + (\overline{\partial} + \psi^{0,1}\wedge) \\ &= \partial_\psi + \overline{\partial}_{\overline{\psi}} \end{aligned}$$

The twisted Dolbeault operators

$$\partial_\psi = \partial + \psi^{1,0}\wedge \tag{3.1}$$

$$\overline{\partial}_{\overline{\psi}} = \overline{\partial} + \psi^{0,1}\wedge \tag{3.2}$$

do not commute i.e. $[\partial_\psi, \overline{\partial}_{\overline{\psi}}] \neq 0$ because of the non-commutative nature of the exterior algebra of the differential forms. Indeed

$$\begin{aligned} \partial_\psi\overline{\partial}_{\overline{\psi}} &= (\partial + \psi^{1,0}\wedge)(\overline{\partial} + \psi^{0,1}\wedge) \\ &= \partial\overline{\partial} + \partial\psi^{0,1} \wedge + \psi^{1,0} \wedge \overline{\partial} + \psi^{1,0} \wedge \psi^{0,1}\wedge \\ &= \partial\overline{\partial} + \psi^{1,0} \wedge \overline{\partial} + \psi^{1,0} \wedge \psi^{0,1}\wedge \\ &= \partial_\psi\overline{\partial} + \psi^{1001}\wedge \end{aligned}$$

Check the above computation, as it is needed in our twisted Ricci form

where $\psi^{1001} = \psi^{1,0} \wedge \psi^{0,1}$ is a $(1,1)$-form. Similarly

$$\overline{\partial}_{\overline{\psi}}\partial_{\psi} = \overline{\partial}_{\overline{\psi}}\partial + \psi^{0110}\wedge$$

To do cohomology with the twisted derivative d_{ψ}, it must hold that $d_{\psi}^2 = 0$

$$\begin{aligned} d_{\psi}^2 &= (\partial_{\psi} + \overline{\partial}_{\overline{\psi}})(\partial_{\psi} + \overline{\partial}_{\overline{\psi}}) \\ &= \partial_{\psi}^2 + \partial_{\psi}\overline{\partial}_{\overline{\psi}} + \overline{\partial}_{\overline{\psi}}\partial_{\psi} + \overline{\partial}_{\overline{\psi}}^2 \\ &= \underbrace{\partial_{\psi}^2 + +\overline{\partial}_{\overline{\psi}}^2}_{?} + \partial_{\psi}\overline{\partial}_{\overline{\psi}} + \overline{\partial}_{\overline{\psi}}\partial_{\psi} \end{aligned}$$

$$\begin{aligned} \partial_{\psi}^2 f &= (\partial + \psi^{1,0}\wedge)(\partial + \psi^{1,0}\wedge)f \\ &= (\partial + \psi^{1,0}\wedge)(\partial f + \psi^{1,0} f) \\ &= \partial^2 f + \partial(\psi^{1,0} f) + \psi^{1,0} \wedge \partial f + \psi^{1,0} \wedge \psi^{1,0} f \\ &= \partial\psi^{1,0} f - \psi^{1,0} \wedge \partial f + \psi^{1,0} \wedge \partial f \\ &= \partial\psi^{1,0} f \neq 0 \end{aligned} \tag{3.3}$$

Q: Why we only consider $\partial_{\psi}\overline{\partial}_{\overline{\psi}} + \overline{\partial}_{\overline{\psi}}\partial_{\psi}$ equal to zero? From 3.3 we see ∂_{ψ}^2 and $\overline{\partial}_{\overline{\psi}}^2$ are also nonzero, so why not consider them here?

from which it also follows that

$$\partial_{\psi}\overline{\partial}_{\overline{\psi}} + \overline{\partial}_{\overline{\psi}}\partial_{\psi} = 0 \tag{3.4}$$

$$\begin{aligned} \partial_{\psi}\overline{\partial}_{\overline{\psi}} + \overline{\partial}_{\overline{\psi}}\partial_{\psi} &= \partial_{\psi}(\overline{\partial} + \psi^{0,1}\wedge) + \overline{\partial}_{\overline{\psi}}(\partial + \psi^{1,0}\wedge) \\ &= \partial_{\psi}\overline{\partial} + \partial_{\psi}\psi^{0,1} \wedge +\overline{\partial}_{\overline{\psi}}\partial + \overline{\partial}_{\overline{\psi}}\psi^{1,0}\wedge \\ &= (\partial_{\psi}\overline{\partial} + \overline{\partial}_{\overline{\psi}}\partial) + (\partial + \psi^{1,0}\wedge)\psi^{0,1} \wedge +(\overline{\partial} + \psi^{0,1}\wedge)\psi^{1,0}\wedge \\ &= (\partial_{\psi}\overline{\partial} + \overline{\partial}_{\overline{\psi}}\partial) + \psi^{1,0} \wedge \psi^{0,1} \wedge +\psi^{0,1} \wedge \psi^{1,0}\wedge \\ &= (\partial_{\psi}\overline{\partial} + \overline{\partial}_{\overline{\psi}}\partial) + \underbrace{\psi^{1,0} \wedge \psi^{0,1} \wedge +\psi^{0,1} \wedge \psi^{1,0}\wedge}_{\psi^{1001}\wedge + \psi^{0110}\wedge} \end{aligned}$$

Q: How does $\psi^{1001} \wedge +\psi^{0110}\wedge$ vanish here?

which is equivalent to

$$\partial_{\psi}\overline{\partial} + \overline{\partial}_{\overline{\psi}}\partial = 0$$

Now we define the subspace of (p,q)- bidegree differential forms

$$\tilde{\Omega}^{p,q}(M) := \{\omega \in \Omega^{p,q}(M) : (\partial_{\psi}\overline{\partial} + \overline{\partial}_{\overline{\psi}}\partial)\omega = 0\} \tag{3.5}$$

as the kernel of the operator $\partial_\psi \overline{\partial} + \overline{\partial}_{\overline{\psi}} \partial$. So our deformed sub complex

$$... \overset{d_\psi^k}{\to} \tilde{\Omega}^k(M) \overset{d_\psi^{k+1}}{\to} \tilde{\Omega}^{k+1}(M) \overset{d_\psi^{k+2}}{\to} \tilde{\Omega}^{k+2}(M) \overset{d_\psi^{k+3}}{\to} ...$$

where $\tilde{\Omega}^k(M) = \bigoplus_{p+q=k} \tilde{\Omega}^{p,q}(M)$, allows us to do cohomology.

There is another remark that we should make about the twisting parameters, that is these parameters should be exact forms. More precisely, the holomorphic one form $\psi^{1,0}$ has the local expression $\psi^{1,0} = f'(z)dz = \partial f$ for some smooth complex valued function f, similarly for the antiholomorphic form we have that $\psi^{0,1} = \overline{\partial} g$ for some other complex valued smooth function g, where f and g are not necessarily the same function.
Then we see that

$$\begin{aligned}
\partial_\psi^2 \omega &= \partial_\psi(\partial\omega + \psi^{1,0} \wedge \omega) \\
&= \partial^2\omega + \psi^{1,0} \wedge \partial\omega + \partial(\psi^{1,0} \wedge \omega) + 0 \\
&= \psi^{1,0} \wedge \partial\omega + \partial\psi^{1,0} \wedge \omega - \psi^{1,0} \wedge \partial\omega \\
&= \partial^2 f \wedge \omega \\
&= 0
\end{aligned}$$

and

$$\begin{aligned}
\overline{\partial}_\psi^2 \omega &= \overline{\partial}_\psi(\overline{\partial}\omega + \psi^{0,1} \wedge \omega) \\
&= \overline{\partial}^2\omega + \psi^{0,1} \wedge \overline{\partial}\omega + \overline{\partial}(\psi^{0,1} \wedge \omega) + 0 \\
&= \psi^{0,1} \wedge \overline{\partial}\omega + \overline{\partial}\psi^{0,1} \wedge \omega - \psi^{0,1} \wedge \overline{\partial}\omega \\
&= \overline{\partial}^2 g \wedge \omega \\
&= 0
\end{aligned}$$

for some $\omega \in \tilde{\Omega}^{p,q}(M)$.

4 Chern classes

We will now attempt to construct first chern class on our complex manifolds. The goal is to build everything in such a manner that if the deformed chern class

$$c_{i,\psi}(TM^{\mathbb{C}}) \in H_\psi^{2i}(M)$$

then putting $\psi = 0$ reduces everything to the classical chern characteristic class theory. This attempt motivates us to deform the Ricci tensor and Ricci form on complex manifolds when we consider the first chern class. Classically, the Ricci form is given as

$$\mathrm{Ric}_g = \sqrt{-1}\partial_i \overline{\partial_j} \log(\det(g_{i\bar{j}}))dz^i \wedge d\bar{z}^j \in \Omega^{1,1}(M) \tag{4.1}$$

that, at *nice* situations, can represent the first chern class i.e.

$$c_1(TM^{\mathbb{C}}) = \frac{\sqrt{-1}}{2\pi}[\mathrm{Ric}_g] \in H^2(M)$$

[1]

So we search for a good candidate for our deformed Ricci form that we shall denote as Ric_g^{ψ} such that

$$c_{1,\psi}(TM^{\mathbb{C}}) = \frac{\sqrt{-1}}{2\pi}[\mathrm{Ric}_g^{\psi}] \in H^2_{\psi}(M) \tag{4.6}$$

Let's try to define the deformed curvature tensor,

$$\mathrm{Ric}_g^{\psi} = \sqrt{-1}\partial_{i,\psi}\overline{\partial_{j,\overline{\psi}}}\log(\det(g_{i\bar{j}})) \in \tilde{\Omega}^{1,1}(M)$$

[2]

5 $\partial\overline{\partial}$-lemmas

We also want a deformed version of the $\partial\overline{\partial}$-lemma saying that on the manifold M if $\alpha \in \tilde{\Omega}^{p,q}(M)$ is d_{ψ}-closed, then locally (i.e. in the neighborhood of any

[1] We may first look for the first Chern class, which has the generalized definition as a map

$$c_1 : \mathrm{Pic}(M) \to H^2(M) \tag{4.2}$$

where the Picard group $\mathrm{Pic}(M)$ is the group of line bundles over M equipped with Hermitian metrics. Pick a member from this group, $L \to M$ which has the Hermitian metric h on it's fibres. Then the curvature is defined as

$$\mathrm{Ric}_h := \sqrt{-1}\partial\overline{\partial}\log(\det(\mathrm{h})) \tag{4.3}$$

We can generalize the map (18) to

$$c_1^{\psi} : \mathrm{Pic}(M) \to H^{1,1}_{\psi}(M) \tag{4.4}$$

with

$$\mathrm{Ric}_h := \sqrt{-1}\partial_{\psi}\overline{\partial_{\overline{\psi}}}\log(\det(\mathrm{h})) \tag{4.5}$$

[2] Let $G = \det(\mathrm{g_{i\bar{j}}})$. We are considering the complex logarithm of this function that is $\log(\mathrm{G})$. Since this is a function $\log(\mathrm{G}) \in \Omega^{0,0}(M)$. Now we apply Dolbeault operators on this complex-valued smooth function.

$$\partial(\log(\mathrm{G})) = \frac{\partial}{\partial z}\log(\mathrm{G})\,dz$$

$$\overline{\partial}(\log(\mathrm{G})) = \frac{\partial}{\partial \overline{z}}\log(\mathrm{G})\,\overline{dz}$$

Experiments:
First apply the twisted Dolbeault machinery in case of an actual Calabi Yau manifold (for example the complex torus), and then there will be some non vanishing terms involving the twisting parameters.
Secondly, apply the the twisted Dolbeault machinery in case of a non-Calabi Yau manifold (for example the complex projective space in dimension n), then it will also have some non-vanishing terms involving the twisting parameters.
We need to compare the non-vanishing terms in both cases.

point of M) we can find a form $\beta \in \tilde{\Omega}^{p-1,q-1}(M)$ such that

$$\alpha = \sqrt{-1}\partial_\psi \overline{\partial}_{\overline{\psi}} \beta$$

so that we can make *deformed* statement similar to the Calabi conjecture. We recall the $\overline{\partial}$-lemma of Poincaré on $\mathbb{C}$. We know that if $f : U \to \mathbb{C}$, where U contains the closure the open ball $B_r(0)$ of radius r centered at the origin, $\overline{B_r(0)}$, is a smooth complex valued function, then it has a correction term in addition to the Cauchy's integral representation formula, that essentially comes from applying Stokes' theorem

$$f(z) = \frac{1}{2\pi i} \int_{\partial B_r(0)} \frac{f(w)}{w - z} dz + \frac{1}{2\pi i} \int_{B_r(0)} \frac{\partial f}{\partial \overline{w}} \frac{dw \wedge d\overline{w}}{w - z}$$

where we clearly see that the holomorphicity of the complex valued function can make the last term vanish, giving us the good old Cauchy's integral representation of holomorphic functions. This representation can be used to show that for any anti holomorphic complex differential form

$$a = f d\overline{z} \in \Omega^{0,1}(U)$$

where is a smooth function as above i.e. $f \in C^\infty(U)$, the function f has anti holomorphic derivative of some other smooth function defined as

$$g(z) := \frac{1}{2\pi i} \int_{B_r(0)} \frac{f(w)}{w - z} dw \wedge d\overline{w} \tag{5.1}$$

So the $\overline{\partial}$-lemma essentially tells us that

$$\frac{\partial g}{\partial \overline{z}} = f$$

[3] Recall the twisted anti holomorphic Dolbeault operator

$$\overline{\partial}_{\overline{\psi}} := \overline{\partial} + \psi^{0,1} d\overline{z}$$

Let us consider the following modified smooth complex valued function: Define the smooth function $\tilde{g}$ such that the following holds true

$$\tilde{g}(z) := \frac{1}{2\pi i} \int_{B_r(0)} \frac{\tilde{f}(w)}{w - z} dw \wedge d\overline{w}$$

where

$$\tilde{f}(z) := (f - \psi^{0,1} \cdot \tilde{g})(z)$$

applying the $\overline{\partial}$-lemma yields

$$\begin{aligned} \frac{\partial \tilde{g}}{\partial \overline{z}} &= \tilde{f} \\ &= f - \psi^{0,1} \cdot \tilde{g} \end{aligned}$$

[3] We won't write the proof this theorem here, as it can be found in any several variable complex analysis textbook, we will just use this theorem

$$\Rightarrow (\frac{\partial \tilde{g}}{\partial \overline{z}} + \psi^{0,1} \cdot \tilde{g}) d\overline{z} = f d\overline{z}$$
$$\Rightarrow \overline{\partial}_{\overline{\psi}} \tilde{g} = a \in \Omega^{0,1}(U)$$

which we may call the Poincaré $\overline{\partial}_{\overline{\psi}}$-lemma.[4]

Example 5.1. Let's apply the twisted Ricci form to the $\mathbb{CP}^1$ with the Fubini-Study metric. The new Dolbeault operators using the closed 1-form $\psi = \psi^{1,0} + \psi^{0,1}$, where:

$$\partial_\psi = \partial + \psi^{1,0}\wedge, \quad \overline{\partial}_{\overline{\psi}} = \overline{\partial} + \psi^{0,1}\wedge$$

Thus, the twisted Ricci form ρ_ψ will be:

$$\rho_\psi = -i\partial_\psi \overline{\partial}_{\overline{\psi}} \log\det(g)$$

Expanding this using the definitions of ∂_ψ and $\overline{\partial}_{\overline{\psi}}$, we get:

$$\rho_\psi = -i\left(\partial + \psi^{1,0}\wedge\right)\left(\overline{\partial} + \psi^{0,1}\wedge\right)\log\det(g)$$

This gives:

$$\rho_\psi = -i\left(\partial\overline{\partial}\log\det(g) + \psi^{1,0}\wedge\overline{\partial}\log\det(g) + \partial\log\det(g)\wedge\psi^{0,1} + \log\det(g)\psi^{1,0}\wedge\psi^{0,1}\wedge\right)$$

$$\rho_\psi = 2i\frac{dz\wedge d\bar{z}}{(1+|z|^2)^2} - i\left(\frac{2\,\psi^{1,0}\wedge d\bar{z}}{1+|z|^2} + \frac{2\,dz\wedge\psi^{0,1}}{1+|z|^2}\right) + i\log\det(g)\psi^{1,0}\wedge\psi^{0,1}\wedge$$

6 Twisted Ricci curvature

Remember we define our connection as,

$$\nabla_{X,\psi}(e_j) = \omega_j^i(X)e_i + \delta_j^i\psi(X)e_i$$

Q: Now, we want to split it like we did for $d = \partial_\psi + \overline{\partial}_{\overline{\psi}}$, but how?

we can write:

$$\nabla = \nabla' + \nabla''$$

The traditional connection ∇ on a complex vector bundle can be expressed in terms of a connection 1-form ω, $\nabla = d + \omega$. We mean that ω encodes the additional structure provided by the connection beyond the trivial connection given by the exterior derivative d. Since a given complex structure on a smooth manifold is the unique splitting of the dual tangent bundle into holomorphic and anti holomorphic components, the decomposition $\Omega^1(M) = \Omega^{1,0}(M) \oplus \Omega^{0,1}(M)$ implies that the connection matrix splits as well i.e. $\omega = \omega' + \omega''$.
Let's define ∇_ψ,

$$\begin{aligned}\nabla_\psi &= \nabla'_\psi + \nabla''_\psi \\ &= \partial_\psi + \omega' + \overline{\partial}_{\overline{\psi}} + \omega''\end{aligned} \tag{6.1}$$

[4] I am a bit uncomfortable about how $\tilde{g}$ is defined. Does such a smooth complex-valued function really exist?

Proposition 6.1.

$$\nabla_{\frac{\partial}{\partial z^i},\psi} e_\alpha = ? \qquad \nabla_{\frac{\partial}{\partial z^i},\psi} \bar{e}_\alpha = ? \qquad \nabla_{\frac{\partial}{\partial \bar{z}^i},\psi} e_\alpha = ? \qquad \nabla_{\frac{\partial}{\partial z^i},\psi} \bar{e}_\alpha = ?$$
$$\nabla_{\frac{\partial}{\partial z^i},\psi} e^\alpha = ? \qquad \nabla_{\frac{\partial}{\partial z^i},\psi} \bar{e}^\alpha = ? \qquad \nabla_{\frac{\partial}{\partial \bar{z}^i},\psi} e^\alpha = ? \qquad \nabla_{\frac{\partial}{\partial z^i},\psi} \bar{e}^\alpha = ?$$

Proof. Writing... □

Useful resources which might be useful:

- About complex conjugate of a vector bundle: `https://math.stackexchange.com/q/3987628/1167703`, `https://en.wikipedia.org/wiki/Complex_conjugate_of_a_vector_space`
- Curvature is $(1,3)$ tensor field: `https://math.stackexchange.com/a/2910771/1167703`

7 Integrable connections and twisted derivatives

It turns out that we can generalize the twisted derivative $d_\psi := d + \psi\wedge$ to integrable connections on flat vector bundles and the cohomology is nothing but the de Rham cohomology of the vector bundle valued differential forms. Given a vector bundle $\pi : E \to M$ of rank r, the connection can be viewed as the map

$$\nabla^U : \Gamma(U, E) \to \Gamma(U, E) \otimes \Omega^1(M) \tag{7.1}$$

for some local trivialization over the open subset $U \subseteq M$. So for local sections $s \in \Gamma(U, E)$

$$\nabla^U(fs) = df \otimes s + f\nabla^U s$$

where $f \in C^\infty(U)$. Choosing a frame $\langle e_1, ..., e_r\rangle$ over U, then we see that since the local sections are generated these e_i's, the connection acts as

$$\nabla^U(e_j) = \sum_{i=1}^{r} e_i \otimes \psi_{ij}$$

where $j = 1, ..., r$ and ψ_{ij} are the so called connection 1-forms. We can define a matrix $\Psi = [\psi_{ij}]$ such that the connection can be expressed as

$$\nabla := d + \Psi \tag{7.2}$$

(from now on we will omit the superscript U on the connection) A connection is called integrable if it squares to zero which is equivalent to the curvature of that bundle (which is defined to be the square of the connection) being zero, such bundles are called flat bundles. So integrability (i.e. $\nabla^2 = 0$) gives us the opportunity to do cohomology for bundle valued differential forms. Recall that

such forms are the tensor product $\Omega^k(M,E) := \Gamma(U, \Lambda^k T^*M \otimes E)$. Flatness of the bundle E gives the cochain complex

$$0 \to \Omega^0(U,E) \xrightarrow{\nabla} \Omega^1(U,E) \xrightarrow{\nabla} \Omega^2(U,E) \xrightarrow{\nabla} \Omega^3(U,E) \to ... \tag{7.3}$$

from which we can extract E-valued de Rham cohomologies $H^k_{dR}(U,E)$. It looks more desirable if the bundle is globally trivial.
Thus, when the bundle is the trivial line bundle i.e. $E = M \times \mathbb{C}$, then the connection matrix Ψ has just a single 1-form ψ, so for trivial line bundles this connection becomes our twisted derivative operator.

8 Cup products and homology

Recall the definition of cap products from algebraic topology

$$\cap : \mathrm{H}_p(M) \times \mathrm{H}^q(M) \to \mathrm{H}_{p-q}(M) \tag{8.1}$$

and that the Poincaré duality theorem of cohomologies can be interpreted using the cap product:

$$\mathrm{H}^i(M) \simeq \mathrm{H}_{n-i}(M) \tag{8.2}$$

via $\alpha \mapsto [M] \cap \alpha$ for any i-form representing the cohomology class $[\alpha] \in \mathrm{H}^i(M)$, the fundamental class $[M]$, provided that the dimension of the compact oriented manifold M is n.
Consider two submanifold N_1, N_2 of M, of codimension i, j, respectively. Suppose further that these submanifolds intersect transversely: $N_1 \cap N_2$, which is a submanifold of M of codimension $i + j$. The corresponding homology classes are

$$[N_1] \in \mathrm{H}_{n-i}(M)$$

$$[N_2] \in \mathrm{H}_{n-j}(M)$$

$$[N_1 \cap N_2] \in \mathrm{H}_{n-i-j}(M)$$

Using (8.2) we deduce that we should have

$$[N_1]^* \in \mathrm{H}^i(M)$$

$$[N_2]^* \in \mathrm{H}^j(M)$$

$$[N_1 \cap N_2]^* \in \mathrm{H}^{i+j}(M)$$

The cup product

$$\smile : \mathrm{H}^i(M) \times \mathrm{H}^j(M) \to \mathrm{H}^{i+j}(M) \tag{8.3}$$

has the interesting property that

$$[N_1]^* \smile [N_1]^* = [N_1 \cap N_2]^* \tag{8.4}$$

It is well known that the twisted cohomology safisfies the Poincaré duality isomorphism, but according to style we have written this duality in (8.2), what should we have on the right hand side? That is,

$$\mathrm{H}^i_\psi(M) \simeq \textbf{(?)} \tag{8.5}$$

What topological implications would we have for cup products of the twisted cohomologies?
Does (8.4) hold for twisted cohomologies?